APPENDIX I
MATHEMATICAL DERIVATIONS

All results are stated for a single action coordinate (continuous) or a single discrete sub-action; the action chunk factorizes over the prediction horizon T_p, the continuous dimensions D_c, and the M discrete sub-actions. Notation follows main-paper Sec. III: continuous belief mean μ and precision ρ, categorical belief $\boldsymbol{\theta} \in \Delta^{K-1}$ (we write K for K_m), and schedule hyperparameters σ_1 (Gaussian) and β_1 (categorical).

A. Conjugate Belief Updates

1) Gaussian–Gaussian (continuous)

Let the prior over a continuous coordinate be $p(a \mid \mu, \rho) = \mathcal{N}(a; \mu, \rho^{-1})$ and a sender message y have likelihood $p(y \mid a; \alpha) = \mathcal{N}(y; a, \alpha^{-1})$. The posterior $p(a \mid y) \propto p(y \mid a)\, p(a)$ has log-density (dropping terms independent of a) latex

$$\begin{aligned} \ln p(a \mid y) &= -\tfrac{\alpha}{2}(y-a)^2 - \tfrac{\rho}{2}(a-\mu)^2 + \text{const} \\ &= -\tfrac{\rho+\alpha}{2}\Big(a - \tfrac{\rho\mu+\alpha y}{\rho+\alpha}\Big)^2 + \text{const}, \end{aligned} \tag{I.1}$$

by completing the square. Hence the posterior is Gaussian with

$$\rho' = \rho + \alpha, \qquad \mu' = \frac{\rho\mu + \alpha y}{\rho + \alpha}, \tag{I.2}$$

which is the continuous update of main-paper Eq. (4): precision adds exactly ($\rho' = \rho + \alpha$) and the mean is a precision-weighted average of prior and message.

2) Categorical in natural parameters (discrete)

A categorical distribution $\boldsymbol{\theta} = \mathrm{softmax}(\boldsymbol{\eta})$ has natural parameters $\boldsymbol{\eta} = \log \boldsymbol{\theta}$ (up to an additive constant). The discrete sender transmits a Gaussian message in this log-probability space,

$$\mathbf{y} \sim \mathcal{N}\big(\alpha(K\mathbf{e}_k - \mathbf{1}),\, \alpha K \mathbf{I}\big), \tag{I.3}$$

whose mean points toward the one-hot target $\mathbf{e}_k$ [17]. Because the message is additive in the natural parameter, the conjugate posterior update is additive in log-space,

$$\log \boldsymbol{\theta}' = \log \boldsymbol{\theta} + \mathbf{y} - \log Z \implies \boldsymbol{\theta}' = \mathrm{softmax}\big(\log \boldsymbol{\theta} + \mathbf{y}\big), \tag{I.4}$$

reproducing the discrete update of main-paper Eq. (4). The update never leaves the simplex, so no relaxation or rounding is needed at any point of sampling.

B. Closed-Form Belief Sampling

Training requires belief states at a random time $t \sim \mathcal{U}(0,1)$ without simulating n sequential updates. Both flows admit an $O(1)$ closed form, which yields main-paper Eq. (2).

1) Continuous

Starting from (μ_0=0, ρ_0=1), after accumulating n messages the belief mean is $\mu_n = (\sum_i \alpha_i y_i)/\rho_n$ with $y_i = x + \epsilon_i/\sqrt{\alpha_i}$. Since the numerator sums independent Gaussians, $\sum_i \alpha_i y_i \sim \mathcal{N}(x \sum_i \alpha_i,\, \sum_i \alpha_i)$, and dividing by ρ_n gives $\mu_n \sim \mathcal{N}(\alpha_t x/\rho_t,\, \alpha_t/\rho_t^2)$ where $\alpha_t = \gamma/(1-\gamma)$ and $\rho_t = 1/(1-\gamma)$. Substituting and simplifying:

$$\mu_t = \gamma(t)\, x + \sqrt{\gamma(t)\big(1-\gamma(t)\big)}\;\epsilon, \qquad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \tag{I.5}$$

with the signal coefficient $\gamma(t) = 1 - \sigma_1^{2t}$ of main-paper Eq. (1). At t=0, γ=0 so μ_0=0 (uninformative prior); at t=1, $\gamma \approx 1$ so $\mu_1 \approx x$.

2) Discrete

By the same marginalization, the training belief at time t is obtained by drawing a single Gaussian message with cumulative accuracy $\beta^{\mathrm{d}}(t) = \beta_1 t^2$ (main-paper Eq. (1)) and applying one softmax:

$$\boldsymbol{\theta}_t = \mathrm{softmax}(\mathbf{y}), \qquad \mathbf{y} \sim \mathcal{N}\big(\beta^{\mathrm{d}}(t)(K\mathbf{e}_k - \mathbf{1}),\, \beta^{\mathrm{d}}(t) K \mathbf{I}\big). \tag{I.6}$$

C. Accuracy Schedules, Conservation, and Step-Count Flexibility

1) Continuous (geometric)

With $\gamma(t) = 1 - \sigma_1^{2t}$ the cumulative precision is $\beta^{\mathrm{c}}(t) = \gamma/(1-\gamma) = \sigma_1^{-2t} - 1$, and the i-th of n steps adds

$$\alpha_i^{\mathrm{c}} = \sigma_1^{-2i/n}\big(1 - \sigma_1^{2/n}\big). \tag{I.7}$$

These telescope to a total independent of n: $\sum_{i=1}^{n} \alpha_i^{\mathrm{c}} = \sigma_1^{-2} - 1 = \beta^{\mathrm{c}}(1)$.

2) Discrete (quadratic)

With $\beta^{\mathrm{d}}(t) = \beta_1 t^2$ the per-step accuracy is $\alpha_i^{\mathrm{d}} = \beta^{\mathrm{d}}(i/n) - \beta^{\mathrm{d}}((i-1)/n) = \beta_1 \frac{2i-1}{n^2}$, and likewise

$$\sum_{i=1}^{n} \alpha_i^{\mathrm{d}} = \frac{\beta_1}{n^2} \sum_{i=1}^{n} (2i-1) = \frac{\beta_1}{n^2}\, n^2 = \beta_1 = \beta^{\mathrm{d}}(1). \tag{I.8}$$

In both flows the *total* information delivered over inference is fixed by the schedule and does not depend on the number of steps n; changing n only repartitions it into finer or coarser increments. The continuous schedule grows exponentially near t=1, concentrating most precision in the final steps, while the discrete schedule distributes precision more evenly through quadratic growth. This reflects the fact that categorical beliefs concentrate more slowly than Gaussian beliefs: even moderate noise obscures class identity, so the discrete flow needs to accumulate evidence more gradually. With β_1 fixed across tasks, the terminal categorical belief also sharpens with K.

3) Validity at any step count

After n updates the continuous belief is $\mathcal{N}(\mu_n, \rho_n^{-1})$ with $\rho_n = 1 + \sum_{i \le n} \alpha_i^{\mathrm{c}}$, and the categorical belief is $\mathrm{softmax}(\log \boldsymbol{\theta}_0 + \sum_{i \le n} \mathbf{y}_i)$. Both are valid posteriors for *every* n, because (i) each conjugate update (App. I-A) is exact regardless of the increment size α_i, (ii) the posterior mean is the Bayes-optimal estimate given the evidence so far, and (iii) the total information $\sum_i \alpha_i$ is conserved across n (above). Moreover, every query time $t_i = (i-1)/n$ lies inside the training distribution $t \sim \mathcal{U}(0,1)$, so reducing n only coarsens the discretization without pushing the network off its training manifold. A diffusion sampler, by contrast, is calibrated to its K-step noise schedule: truncating it leaves the sample at a noise level the network was never trained to denoise from. Empirically, one trained BFN Policy checkpoint is stable from n=50 down to $n \approx 10$ and collapses only at n=1 (main-paper Fig. 5; protocol in App. IV-E).

D. Training Objective: From Sender–Receiver KL to the Practical Loss

The BFN objective minimizes the Kullback–Leibler (KL) divergence between the sender distribution (centered at the expert target) and the receiver distribution (centered at the network prediction) [17]. We give the per-step form for each flow; the motivation for the practical surrogates is discussed in main-paper Sec. III-B, so we keep it brief here.

1) Continuous

For sender $\mathcal{N}(x, \alpha^{-1})$ and receiver $\mathcal{N}(\hat{x}, \alpha^{-1})$,

$$D_{\mathrm{KL}} = \tfrac{\alpha}{2}\,(\hat{x} - x)^2, \tag{I.9}$$

so the exact time-continuous loss carries weight $\alpha(t) = \gamma/(1-\gamma) = \sigma_1^{-2t} - 1$, which *diverges* as $t \to 1$ ($\sim 10^6$ for $\sigma_1 = 10^{-3}$) and destabilizes training. We therefore optimize the bounded $\gamma(t)$-weighted squared error of main-paper Eq. (3),

$$\mathcal{L}_{\text{cont}} = \mathbb{E}_{t,\epsilon}\big[\gamma(t)\,\|\mathbf{x} - \Phi_{\text{cont}}(\boldsymbol{\mu}_t, \boldsymbol{\theta}_t, t, \mathbf{o})\|_2^2\big], \qquad \gamma(t) \in [0,1], \tag{I.10}$$

which retains the emphasis on the low-noise (high-γ) regime that dominates the final sample while remaining numerically stable, in the spirit of the "simple" loss of [23]. (The continuous-only Push-T experiments of main-paper Sec. IV-B drop the weighting and use plain MSE, matching the Diffusion Policy objective exactly.)

2) Discrete

For the categorical sender/receiver of Eq. (I.3), the sender and receiver share covariance $\alpha K \mathbf{I}$ and their means differ only through $\mathbf{e}_k$ vs. the predicted class probabilities $\hat{\mathbf{p}}$, so the divergence reduces to a precision-weighted squared error:

$$D_{\mathrm{KL}} = \tfrac{1}{2}(\boldsymbol{\mu}_S - \boldsymbol{\mu}_R)^\top (\alpha K \mathbf{I})^{-1} (\boldsymbol{\mu}_S - \boldsymbol{\mu}_R) = \tfrac{\alpha K}{2}\|\mathbf{e}_k - \hat{\mathbf{p}}\|_2^2. \tag{I.11}$$

Squared error on softmax outputs has vanishing gradients once predictions saturate, so the discrete flow optimizes the cross-entropy surrogate of main-paper Eq. (3), which shares the same minimizer ($\hat{\mathbf{p}} = \mathbf{e}_k$) but provides well-scaled gradients:

$$\mathcal{L}_{\text{disc}} = \mathbb{E}_{t,\mathbf{y}} \sum_{m=1}^{M} \mathrm{CE}\big(\Phi_{\text{disc}}^{(m)}(\boldsymbol{\mu}_t, \boldsymbol{\theta}_t, t, \mathbf{o}),\, k^{(m)}\big). \tag{I.12}$$

The full objective is $\mathcal{L} = \mathcal{L}_{\text{cont}} + \mathcal{L}_{\text{disc}}$, used by main-paper Alg. 1.

Appendix II
Network Architecture and Implementation Details

A. Shared Backbone and Encoders

All methods share an identical backbone and observation encoder, differing only in their generative formulation and inference procedure; this isolates the action representation as the only variable across methods. The denoising/belief network is the Conditional 1D U-Net of Diffusion Policy [7], with down-sampling channel dimensions $[256, 512, 1024]$ for the PAMDP and real-robot experiments and $[512, 1024, 2048]$ for the image-based continuous Push-T simulation, convolution kernel size 5, 8 groups per group normalization, and feature-wise linear modulation (FiLM) for conditioning. A 128-dimensional sinusoidal embedding of the diffusion/flow time is concatenated with the encoded observation features and injected into every residual block. State-based PAMDP tasks use a two-layer MLP observation encoder with hidden sizes $[256, 256]$; image-based tasks use a ResNet-18 backbone with group normalization and a spatial-softmax pooling head. The network operates on the action chunk over the prediction horizon $T_p = 16$.

B. Input/Output Projections

For the diffusion baselines the per-step channel width equals the action dimension, where each discrete sub-action is encoded as a one-hot vector and treated as a continuous variable. For BFN Policy the channel width is $D_c + \sum_{m=1}^{M} K_m$: the continuous belief mean is concatenated with the per-sub-action categorical beliefs, and the network output is split back into a continuous prediction and M blocks of class logits (main-paper Sec. III-C). The two formulations therefore use the same backbone with identical input and output projection widths; only what the tensor represents (beliefs vs. noisy samples) and what the network predicts (clean action and logits vs. noise) differ. The full BFN Policy training and sampling procedures are given in main-paper Algs. 1 and 2.

C. Diffusion-Baseline Handling of Discrete Actions

On PAMDP tasks the diffusion baselines follow the standard embed–diffuse–round recipe [50], [51], [52]: each discrete sub-action is one-hot encoded, corrupted with Gaussian noise jointly with the continuous dimensions, and recovered at the end of sampling by an argmax over the one-hot block. Relaxing the simplex in this way breaks the discrete–continuous coupling that the Gumbel-softmax line of work also documents; the diagnostics of App. III measure this directly. DDPM uses 100 denoising steps [23]; DDIM runs 10 steps from the pretrained DDPM checkpoint [23]; Consistency Policy [15] is distilled from the same DDPM checkpoint and evaluated with 1-step and 3-step inference.

D. Timing Protocol

All inference times are measured per predicted action chunk at batch size 1 on 2 NVIDIA RTX A6000 GPUs, reported as mean ± 95% CI over rollout episodes. NFE counts network forward passes per action chunk and is hardware-independent (main-paper Sec. IV-A); BFN-n performs n Bayesian updates plus one read-out pass at $t{=}1$, i.e., $n{+}1$ evaluations, and all reported NFE include the read-out.

Appendix III
Sampler Diagnostic Protocol (Main-Paper Fig. 1(c))

The diagnostics are run on Hard Move with $n_{\text{act}} \in \{4, 6, 8\}$ actuators ($K = 2^{n_{\text{act}}} \in \{16, 64, 256\}$ discrete subsets; one signed continuous magnitude). They probe the

internal state of each sampler: at every inference step we ask what discrete action the policy would commit to if forced to stop, and how its continuous prediction relates to that commitment.

A. Committed Discrete Decode at Step i

For BFN Policy, the committed discrete action at step i is $\hat{d}_i = \arg\max_k \ell_i^{(m)}$, the argmax of the class logits produced by the network pass at $t_i = (i-1)/n$. For the diffusion baselines, the noisy sample at step i is first converted to the standard clean-action estimate $\hat{\mathbf{x}}_0$ (from the predicted noise $\hat{\epsilon}$), and the committed discrete action is the argmax over the one-hot channel block of $\hat{\mathbf{x}}_0$.

B. The Three Rows of Fig. 1(c)

- **Top (discrete accuracy along sampling):** the fraction of states at which $\hat{d}_i$ equals the expert's displacement direction d^*, plotted against the step fraction i/n. The dashed line scores the DLPA expert's own actions against the demonstrations; it is below 100% because the demonstrations are multimodal, and it upper-bounds what any imitator can reach.
- **Middle (magnitude error when right):** the final continuous error $|\hat{x} - x^*|$ averaged over the states where the final committed discrete action is correct.
- **Bottom (coupling):** the relative reduction $(E_{\text{wrong}} - E_{\text{right}})/E_{\text{wrong}}$, where E_{right} and E_{wrong} are the magnitude errors conditioned on the discrete action being right or wrong. A value of 0 means the magnitude is predicted independently of the discrete choice; the expert's 55–76% sets the reference level of coupling present in the data.

All curves and bars are reported with 95% CIs.

APPENDIX IV
PAMDP EXPERIMENTS (Q1)

A. Environment Descriptions

We evaluate on eight PAMDP benchmarks: Platform and Goal [10], Catch Point [36], and Hard Goal and four Hard Move variants [37]. All use low-dimensional state observations and a parameterized action space in which the agent jointly selects a discrete action type and its associated continuous parameters. Table IV.I summarizes the action and observation structure of each benchmark.

- **Platform**: The agent must navigate to the goal while avoiding enemies and gaps. It selects a discrete locomotion type (run, hop, leap) along with a continuous horizontal displacement. The observation is 9-dimensional and the maximum episode length is 250.
- **Goal**: The agent aims to score by shooting the ball into the gate. Three parameterized actions are available: kick-to(x, y), shoot-goal-left(h), and shoot-goal-right(h), where the continuous parameters specify a 2-D position or a location along the goal line (padded to two continuous dimensions).
- **Hard Goal**: A more challenging variant that divides the shooting action into ten parameterized actions by evenly partitioning the goal line, with each action's continuous parameter mapped to its corresponding region (11 discrete actions in total). The observation is 17-dimensional and the maximum episode length is 100.
- **Catch Point**: The agent must reach and grab a target point. Two parameterized actions are available: *move*, parameterized by a continuous directional value, and *catch*, which attempts to grab the target. The observation is 6-dimensional and the maximum episode length is 50.
- **Hard Move**: The agent controls n_{act} equally spaced actuators to reach a target area. Each actuator can be toggled on or off, resulting in $2^{n_{\text{act}}}$ discrete action combinations, and a continuous magnitude scales the resulting composite displacement. As n_{act} grows, the discrete action space scales exponentially ($16 \rightarrow 1024$ for $n_{\text{act}} \in \{4, 6, 8, 10\}$) while the continuous parameterization is held fixed, isolating the effect of discrete cardinality. The observation is 4-dimensional and the maximum episode length is 25.

B. Expert Policy and Demonstrations

Across all eight tasks we use DLPA [38], a state-of-the-art model-based RL algorithm for PAMDPs, as the expert from which demonstrations are collected, trained with the authors' default hyperparameters (Table IV.II). Only successful trajectories are retained. The demonstration budget is task-adaptive rather than fixed, since tasks differ in trajectory diversity: more constrained tasks admit fewer distinct successful behaviors and therefore need fewer demonstrations for sufficient behavioral coverage. The resulting per-task counts (132–494) are listed in the "# Demos" column of main-paper Table II. Demonstrations are sliced into length-16 windows with one step of pre-padding and seven of post-padding, with a 2% held-out validation split (seed 42).

C. Policy Hyperparameters

All policies share the chunking, backbone, observation encoder, optimizer, and training budget in Table IV.III, differing only in the generative formulation and the number of inference steps. BFN Policy uses $\sigma_1 = 10^{-3}$ for the Gaussian flow and $\beta_1 = 0.2$ for the categorical flow, with $n = 20$ sampling steps; because the belief remains a valid posterior throughout inference (App. I-C.3), the *same* trained model is also run with $n = 10$ (**BFN-10**) without distillation or retraining. Among the diffusion baselines, DDPM uses 100 denoising steps; DDIM uses 10 steps from the pretrained DDPM checkpoint; and Consistency Policy is distilled from the same checkpoint and evaluated with 1-step and 3-step inference (App. II-C).

D. Cumulative-Reward Results

Main-paper Table II reports success rates; Table IV.IV complements it with cumulative episode reward under the identical protocol (50 episodes, mean $\pm$ 95% CI). The reward metric reproduces the success-rate conclusions: BFN Policy achieves the highest or near-highest reward among learned policies on all tasks, the margin over diffusion and consistency baselines grows with discrete cardinality, and BFN-10 matches BFN-20 within CI throughout.

Benchmark	Obs. dim	# Discrete	Cont. dim	Max steps	Success criterion
Hard Move (n_{act}=4)	4	16	1	25	$R_{\text{ep}} > 0$
Hard Move (n_{act}=6)	4	64	1	25	$R_{\text{ep}} > 0$
Hard Move (n_{act}=8)	4	256	1	25	$R_{\text{ep}} > 0$
Hard Move (n_{act}=10)	4	1024	1	25	$R_{\text{ep}} > 0$
Catch Point	6	2	1	50	$R_{\text{ep}} > 0$
Goal	17	3	2	100	$R_{\text{ep}} \geq 40$
Hard Goal	17	11	2	100	$R_{\text{ep}} \geq 40$
Platform	9	3	1	250	$R_{\text{ep}} > 0.9$

TABLE IV.I: Specification of the eight PAMDP benchmarks. "# Discrete" counts the discrete sub-action combinations; "Cont. dim" is the dimensionality of the continuous parameters (padded across action types where they differ); "Max steps" is the episode-length cap at which a rollout ends; R_{ep} denotes cumulative episode reward.

Hyperparameter	Value
Env. steps (max)	5×10^5
Model type / hidden width	multi-head / 64
Optimizer / learning rate	Adam / 3×10^{-4}
Batch size / updates per step	64 / 25
Seed steps / replay buffer	50 / 10^6
MPC horizon / discount γ	8 / 0.99
CEM pop. / elites / iters	1000 / 100 / 6
CEM temp. / momentum α	0.5 / 0.1
Min. action std / policy mix.	0.05 / 0.05
Loss coef. (cons./rew./cont./val.)	2.0 / 0.5 / 0.5 / 0.1
PER α / β	0.6 / 0.4
Grad. clip norm	10

TABLE IV.II: DLPA expert hyperparameters used to collect demonstrations on all PAMDP benchmarks.

Hyperparameter	Value
Observation horizon T_o	2
Prediction horizon T_p	16
Action (execution) horizon T_a	8
U-Net down dims	$[256, 512, 1024]$
Kernel size / groups	5 / 8
Time embedding dim	128
Obs. encoder (state)	MLP $[256, 256]$
Optimizer	AdamW
Learning rate	1×10^{-4}
Betas / weight decay / eps	$(0.95, 0.999)$ / 10^{-6} / 10^{-8}
LR schedule / warmup	cosine / 500 steps
Epochs	200
Batch size	128
EMA power / max decay	0.75 / 0.9999
Validation ratio	0.02
BFN Policy / BFN Policy-10 steps	20 / 10
BFN Policy: σ_1, β_1	10^{-3}, 0.2
DDPM / DDIM steps	100 / 10
Consistency Policy steps	1, 3

TABLE IV.III: Training and inference hyperparameters shared by BFN Policy and the diffusion-based baselines on the PAMDP benchmarks.

E. Step-Count Ablation Protocol

BFN Policy runs a single trained checkpoint at any number of inference steps n without retraining or distillation (App. I-C.3). Main-paper Fig. 5 sweeps $n \in \{1, 2, 4, 8, 10, 20, 30, 40, 50\}$ for one fixed checkpoint per environment, reporting success rate and cumulative reward over 50 episodes per setting. Accuracy rises steeply up to $n \approx 8$–10 and is then essentially flat through n=50: reducing the step count from 50 to 10 leaves performance unchanged, and only the extreme n=1 setting degrades. This confirms empirically that the belief is a valid posterior at every step count, so test-time compute can be tuned to the latency budget with no loss in quality down to $n \approx 10$.

APPENDIX V
CONTINUOUS PUSH-T EXPERIMENTS (Q2)

A. Simulation Setup

We use the standard simulated, image-based Push-T benchmark of Diffusion Policy [7], whose action is a purely continuous 2-D target position (no discrete component, so BFN Policy reduces to the Gaussian Bayesian flow and, dropping the γ-weighting as noted in App. I-D, shares the Diffusion Policy objective exactly). The policy is conditioned on a $3 \times 96 \times 96$ RGB image and the 2-D agent position over the last $T_o = 2$ frames, encoded with the ResNet-18 / spatial-softmax encoder and an 84×84 center crop. We train on the standard 206-demonstration dataset for 300 epochs with batch size 64, widening the U-Net to $[512, 1024, 2048]$ and keeping all other settings as in App. IV-C; BFN Policy uses $n = 20$ steps. Evaluation uses 50 test rollouts (episodes end on success or at 300 steps), reporting max target coverage. Table V.I lists the settings of both Q2 variants; results are in main-paper Table III.

B. WidowX Hardware Setup

To verify that BFN Policy's belief-based sampler stays competitive when the action space is *purely continuous* on real hardware, a closed-gripper WidowX arm performs the same Push-T task (main-paper Fig. 4(b)). Visual observations come from a front camera and a side camera at $3 \times 512 \times 512$ resolution; the IoU metric is computed from the side view only, as the last-frame intersection-over-union against the goal pose. We train on 150 demonstrations collected by tele-operation at 20 Hz and evaluate $N = 15$ rollouts per method; all other settings are unchanged from simulation, and all methods share backbone and encoders. Quantitative results, including per-call inference time and NFE, are in main-paper Table III: BFN Policy-20 attains the highest mean IoU (0.88), above DDIM-10 (0.85) and DDPM-100 (0.82) at 5× less compute than the latter, while the consistency baselines lose roughly half their simulation performance, confirming that BFN Policy's advantage is not limited to hybrid action spaces.

Task	DLPA (Expert)	Consistency-1	Consistency-3	DDPM-100	DDIM-10	BFN-10	BFN-20
Platform	0.99 ± 0.01	0.14 ± 0.00	0.17 ± 0.02	1.00 ± 0.01	**1.00 ± 0.00**	**1.00 ± 0.00**	0.99 ± 0.02
Catch Point	8.46 ± 1.22	-58.45 ± 9.04	-27.84 ± 5.64	9.30 ± 2.91	7.77 ± 3.00	**10.25 ± 2.55**	10.20 ± 2.55
Goal	18.52 ± 2.59	-10.56 ± 0.40	-9.97 ± 0.99	2.56 ± 7.08	-7.48 ± 4.80	14.87 ± 8.12	**17.34 ± 8.14**
Hard Goal	17.75 ± 2.56	-9.14 ± 0.41	-8.30 ± 0.56	0.49 ± 6.92	-2.12 ± 5.94	**2.32 ± 7.06**	2.04 ± 7.16
Hard Move (n_{act}=4)	3.40 ± 1.79	-26.52 ± 7.54	-30.34 ± 7.50	-1.48 ± 5.82	-5.09 ± 6.64	3.79 ± 5.05	**3.95 ± 5.04**
Hard Move (n_{act}=6)	0.32 ± 2.45	-30.48 ± 6.58	-30.96 ± 6.43	-14.73 ± 5.21	-18.50 ± 5.71	**4.34 ± 4.93**	4.20 ± 5.07
Hard Move (n_{act}=8)	-3.69 ± 2.27	-31.26 ± 6.63	-31.22 ± 6.54	-31.26 ± 6.89	-31.17 ± 6.73	-3.05 ± 6.15	**-2.44 ± 6.13**
Hard Move (n_{act}=10)	-19.16 ± 5.56	-31.56 ± 6.80	-31.08 ± 6.78	-31.44 ± 6.72	-31.23 ± 6.97	-20.50 ± 7.10	**-19.95 ± 7.13**

TABLE IV.IV: **Cumulative reward on PAMDP benchmarks.** Mean ± 95% CI over 50 episodes, under the protocol of main-paper Table II (which reports success rates on the same runs). The suffix in each method name denotes its number of inference steps; DLPA is the RL expert used for demonstration collection. Best IL result per task in **bold**.

Hyperparameter	Simulation	WidowX (real)
Observation & action		
Action	2-D target position	
Cameras	1	2 (front + side)
Image res. / eval crop	96^2 / 84^2	512^2 / 384^2
Low-dim. observation	2-D agent position	—
Model		
Image encoder	ResNet-18 + spatial softmax (GN)	
U-Net down dims	$[512, 1024, 2048]$	
T_o / T_p / T_a	2 / 16 / 8	
Training		
Demonstrations	206	150 (teleop, 20 Hz)
Epochs	300	200
Batch size / val. ratio	64 / 0.02	
Loss	plain MSE (no γ-weighting)	
Inference & evaluation		
BFN Policy: σ_1 / steps	10^{-3} / 20	
DDPM / DDIM / Consistency steps	100 / 10 / $\{1, 3\}$	
Rollouts	50	15
Metric	max coverage	last-frame IoU

TABLE V.I: Hyperparameters for the continuous Push-T experiments (Q2) in simulation and on the WidowX arm. Images are RGB (3 channels). All methods share the backbone and training protocol, differing only in the generative formulation and number of inference steps.

Hyperparameter	Hybrid Push-T (xArm)
Action / # disc. / cont. dim	($d \in \{0, \dots, 7\}$, x) / 8 / 1
Cameras (policy)	1 (front)
Image res. / eval crop	3×224×224 / 216×216
Image encoder	ResNet-18 + spatial softmax (GN)
U-Net down dims	$[256, 512, 1024]$
T_o / T_p / T_a	2 / 16 / 8
BFN Policy: σ_1, β_1, steps	10^{-3}, 0.2, 20
Epochs / batch size	200 / 32
Demos / val. ratio	200 (teleop, 30 Hz) / 0.10
Rollouts / step cap	20 / 150

TABLE VI.I: Hyperparameters for the real-world hybrid Push-T experiments on the xArm. All other settings match Table IV.III.

APPENDIX VI
REAL HYBRID PUSH-T EXPERIMENTS (Q3)

A. Task and Hybrid Action Space

The primary real-world task is Push-T on a physical xArm manipulator: the robot must push a T-shaped block from randomized initial configurations into a fixed goal pose, with the end-effector operating in the image plane. Each action is the hybrid pair $\mathbf{a} = (d, x)$ with a discrete push direction $d \in \{0, \dots, 7\}$ and a continuous push distance x. The eight directions are the cardinal and diagonal unit vectors $\mathbf{u}_d$ (diagonals normalized to unit norm), and the executed end-effector displacement is $\Delta = \mathbf{u}_d\, x$. This factorization mirrors how a human teleoperator pushes—commit to a direction, then choose how far—and makes the discrete–continuous coupling explicit: the same x is correct or catastrophic depending on d, the structure that relaxation-based continuous policies cannot represent natively.

The policy is conditioned on a single front RGB camera at $3 \times 224 \times 224$ resolution over the last $T_o = 2$ frames (center-cropped to 216×216 at evaluation); a top-down camera is used only to compute the evaluation IoU (main-paper Fig. 6). Demonstrations are collected by teleoperation at 30 Hz and converted to the hybrid `(direction, distance)` representation ; we use 200 demonstration episodes with a 10% validation split.

B. Evaluation Protocol and Metric

We run 20 rollouts per method from matched initial block poses, each capped at 150 steps, and measure block–goal alignment as the mask IoU from the top-down camera. We report the maximum of a 5-frame moving average of the per-frame IoU after discarding the first 20% of frames, which rewards reaching *and holding* alignment, together with NFE, per-call inference time, and episode duration (main-paper Table IV). All values are mean ± 95% CI over the 20 rollouts.

C. Policy Hyperparameters

All policies share the ResNet-18 spatial-softmax encoder, backbone, BFN Policy schedule, optimizer, and chunking of the simulation experiments (Table IV.III), differing only in the generative head and inference steps; Table VI.I lists the settings specific to the hybrid real-robot policy.